

Highlights

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Huanjie Yu

- DVCL introduces dual-view contrastive learning that aligns graph-structural and fingerprint-chemical views of molecules
- Chemistry-aware augmentations (atom masking, edge dropping, subgraph extraction) preserve molecular semantics during pre-training
- Pre-training with the dual-view InfoNCE objective improves AUROC by 1.4–2.4 points on MoleculeNet benchmarks
- Concatenated graph+fingerprint fine-tuning leverages complementary information from both views for downstream prediction
- Ablation studies validate that dual-view design outperforms single-view contrastive alternatives

DVCL: Dual-View Contrastive Learning for Molecular Property Prediction on MoleculeNet*

Huanjie Yu^{a,*,1}

^a*School of Computer Science, University Road, Beijing, 100000, Beijing, China*

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ABSTRACT

Molecular property prediction is a fundamental task in drug discovery, yet labeled biochemical data remain scarce and expensive to obtain. Self-supervised contrastive learning has emerged as a promising paradigm for leveraging unlabeled molecular data, but existing approaches construct two augmented views from the same graph-based representation, limiting the diversity of learned features. In this paper, we propose DVCL (Dual-View Contrastive Learning), a self-supervised pre-training framework that contrasts two *inherently different* molecular representations: a graph-structural view captured by a Directed Message Passing Neural Network (DMPNN) and a chemical-fingerprint view captured by a multi-layer perceptron (MLP) operating on 2048-dimensional Morgan count fingerprints. These two views encode complementary aspects of molecular identity—topological connectivity and circular substructure composition—and their alignment through an InfoNCE contrastive objective produces richer, more transferable representations than single-view or same-representation contrastive methods. Chemistry-aware augmentations, including atom masking, edge dropping, and subgraph extraction, are applied to the graph view to preserve molecular semantics during pre-training. Upon fine-tuning, the graph and fingerprint embeddings are concatenated and passed through a task-specific classifier. We evaluate DVCL on four MoleculeNet benchmarks—BBBP, BACE, HIV, and Tox21—using scaffold splitting. DVCL with pre-training achieves AUROC scores of 0.9245 on BBBP and 0.8875 on BACE, substantially outperforming standard DMPNN without pre-training (0.9102 and 0.8636, respectively) and competitive GNN baselines. Ablation studies confirm that both the dual-view design and the chemistry-aware augmentations contribute independently to the observed gains. Our results demonstrate that contrasting structurally and chemically distinct molecular views is a simple yet effective strategy for self-supervised molecular representation learning.

1. Introduction

Molecular property prediction is a central task in computational drug discovery, enabling researchers to estimate the biological activity, toxicity, and pharmacokinetic profiles of candidate compounds before committing to costly wet-laboratory experiments (Wu, Ramsundar, Feinberg, Gomes, Geniesse, Pappu, Leswing and Pande, 2018). Accurate *in silico* screening can dramatically accelerate the drug development pipeline by prioritizing promising molecules from vast chemical libraries. However, a persistent bottleneck in building effective prediction models is the scarcity of labeled biochemical data: experimental assays are expensive, time-consuming, and often yield only hundreds to low thousands of labeled molecules for a given endpoint (Yang, Swanson, Jin, Coley, Eiden, Gao, Guzman-Perez, Hopper, Kelley, Mathea et al., 2019).

Graph neural networks (GNNs) have emerged as the dominant paradigm for molecular property prediction (Wu, Pan, Chen, Long, Zhang and Yu, 2021; Zhang, Chen, Li, Pan, Chen, Li and Chen, 2020). By representing molecules as graphs—atoms as nodes and chemical bonds as edges—GNNs learn hierarchical molecular representations through iterative message passing (Gilmer, Schoenholz, Riley, Vinyals and Dahl, 2017). Architectures such as GCN (Kipf and Welling, 2017), GAT (Veličković, Cucurull, Casanova, Romero, Liò and Bengio, 2018), GIN (Xu, Hu, Leskovec and Jegelka, 2019), and the Directed Message Passing Neural

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*Corresponding author. E-mail: 2990280345@qq.com

 <https://orcid.org/0009-0001-4664-1119>

ORCID(s):

Network (DMPNN) (Yang et al., 2019) have demonstrated strong performance on standardized benchmarks. Nevertheless, these supervised approaches are constrained by the limited availability of labeled data, and their representations can overfit to the specific training distribution.

To address the labeled-data bottleneck, self-supervised pre-training has attracted growing interest in the molecular machine learning community (Hu, Liu, Gomes, Zitnik, Liang, Pande and Leskovec, 2020; Rong, Bian, Xu, Xie, Wei, Huang and Huang, 2020). Inspired by the success of contrastive learning in computer vision (Chen, Kornblith, Norber and Hinton, 2020; He, Fan, Wu, Xie and Girshick, 2020) and natural language processing (Gao, Yao and Chen, 2021), several recent works have adapted contrastive objectives to molecular graphs (Wang, Guo, Wang, Sun and Huang, 2022; You, Chen, Sui, Chen, Wang and Shen, 2020). The standard recipe involves constructing two augmented views of the *same* molecular graph representation and training the encoder to produce similar embeddings for positive pairs while pushing apart negative pairs. Augmentation strategies typically include atom masking, bond deletion, and subgraph removal.

However, a fundamental limitation of existing molecular contrastive learning methods is that both views are derived from the same underlying representation—the molecular graph. This means that the two augmented views share the same encoding paradigm (*i.e.*, graph neural message passing) and differ only in the specific perturbations applied. Consequently, the contrastive signal is limited to teaching the encoder to be invariant to graph-level corruptions, rather than learning to align fundamentally different *perspectives* on molecular identity.

In this paper, we propose **DVCL** (**D**ual-**V**iew **C**ontrastive **L**earning), a self-supervised pre-training framework that addresses this limitation by contrasting two *inherently different* molecular representations. The key insight is that molecular identity can be captured from at least two complementary perspectives:

1. **Graph-structural view:** A graph neural network (specifically, DMPNN) encodes the topological structure of the molecule, capturing atom-level connectivity patterns, bond types, and multi-hop neighborhood relationships through directed message passing.
2. **Fingerprint-chemical view:** A multi-layer perceptron (MLP) encodes a 2048-dimensional Morgan count fingerprint, which represents the molecule as a vector of circular substructure counts. Morgan fingerprints encode fundamentally different information from graph representations: they aggregate substructure frequencies without preserving explicit topological connectivity.

By training these two encoders to produce aligned embeddings through an InfoNCE contrastive loss (Oord, Li and Vinyals, 2018), DVCL forces both views to agree on molecular identity while preserving view-specific information. This dual-view design provides a richer learning signal than same-representation contrastive learning because the two views encode complementary aspects of molecular structure: the graph view captures global topology and long-range atom interactions, while the fingerprint view captures local substructure composition and frequency patterns.

During fine-tuning, the pre-trained graph and fingerprint embeddings are concatenated and passed through a task-specific MLP classifier. This simple fusion strategy allows the downstream model to leverage both topological and chemical features, benefiting from the complementary information captured during pre-training.

We evaluate DVCL on four MoleculeNet benchmarks (Wu et al., 2018)—BBBP, BACE, HIV, and Tox21—using scaffold splitting to simulate realistic drug-discovery evaluation. Our experiments yield the following key findings:

- DVCL with dual-view pre-training achieves AUROC scores of 0.9245 on BBBP and 0.8875 on BACE, outperforming the same architecture without pre-training (0.9102 and 0.8636, respectively) and standard GNN baselines.
- The dual-view design consistently outperforms single-view contrastive alternatives that use only the graph encoder with augmented molecular views.
- Chemistry-aware augmentations (atom masking, edge dropping, subgraph extraction) outperform random augmentations for molecular contrastive pre-training.
- The concatenated graph+fingerprint representation provides complementary information that neither view alone can fully capture.

The remainder of this paper is organized as follows. Section 2 reviews related work on GNNs for molecules, self-supervised contrastive learning, and molecular feature engineering. Section 3 describes the DVCL architecture, augmentation strategies, and training procedure. Section 4 presents the experimental setup, results, and ablation studies. Section 5 discusses the implications of our findings and the advantages of dual-view contrastive learning. Finally, Section 6 concludes with a summary and future directions.

2. Related Work

2.1. Graph Neural Networks for Molecular Property Prediction

Graph neural networks have become the standard approach for molecular property prediction (Wu et al., 2021). The foundational work by Duvenaud, Maclaurin, Iparraguirre, Bombarell, Hirzel, Aspuru-Guzik and Adams (2015) introduced neural graph fingerprints, which generalize traditional molecular fingerprints through differentiable message passing. Gilmer et al. (2017) proposed the Message Passing Neural Network (MPNN) framework, unifying various GNN architectures under a common formulation where messages are computed from neighboring node features, aggregated, and used to update node representations.

Several specific GNN architectures have been widely adopted for molecular tasks. GCNs (Kipf and Welling, 2017) extend convolutional operations to graphs by aggregating normalized neighbor features. GATs (Veličković et al., 2018) introduce attention mechanisms to weigh neighbor importance during aggregation. GINs (Xu et al., 2019) provide theoretical analysis of GNN expressiveness and achieve maximal discriminative power among message passing architectures. GraphSAGE (Hamilton, Ying and Leskovec, 2017) introduces sampling-based neighborhood aggregation for inductive learning on large graphs.

The Directed Message Passing Neural Network (DMPNN) of Yang et al. (2019) addresses a key limitation of standard GNNs: informational redundancy from undirected message passing. By associating messages with directed edges rather than nodes, DMPNN ensures that each message carries genuinely new information. The architecture uses bond features to directly modulate information flow and achieves strong performance on MoleculeNet benchmarks. We adopt DMPNN as the graph encoder in DVCL due to its proven effectiveness and its ability to incorporate bond-level chemical information.

2.2. Self-Supervised and Contrastive Learning for Molecules

Self-supervised learning aims to learn useful representations from unlabeled data by solving pretext tasks, reducing dependence on expensive labeled datasets (Liu, Guo and Tang, 2022). For molecular graphs, self-supervised approaches can be broadly categorized into generative and contrastive methods.

Generative methods learn representations by reconstructing masked or corrupted molecular components. Rong et al. (2020) proposed GROVER, which pre-trains a graph transformer on large-scale molecular data using node/edge-level and graph-level self-supervised tasks. Hu et al. (2020) introduced strategies for pre-training GNNs at both the node level (attribute masking, context prediction) and graph level (similarity prediction), demonstrating that pre-trained representations transfer effectively to downstream property prediction tasks.

Contrastive methods learn representations by maximizing agreement between differently augmented views of the same data. In computer vision, SimCLR (Chen et al., 2020) and MoCo (He et al., 2020) demonstrated that simple contrastive objectives can match or exceed supervised pre-training. For molecules, You et al. (2020) proposed GraphCL, which applies graph-level augmentations (node dropping, edge perturbation, subgraph sampling, attribute masking) and uses an InfoNCE loss to learn augmentation-invariant representations. Wang et al. (2022) introduced MolCLR, which applies similar augmentations at the molecular level with a focus on chemically meaningful perturbations.

Despite their success, these molecular contrastive learning methods share a common limitation: both positive views are derived from the same graph representation, differing only in the augmentations applied. The contrastive signal thus teaches augmentation invariance rather than cross-representation alignment. DVCL addresses this by contrasting fundamentally different representations—graph structure and molecular fingerprints—which encode complementary information about molecular identity.

2.3. Molecular Fingerprint Representations

Molecular fingerprints are fixed-length vector representations that encode molecular structure for machine learning and similarity searching. Morgan/ECFP fingerprints (Rogers and Hahn, 2010) are among the

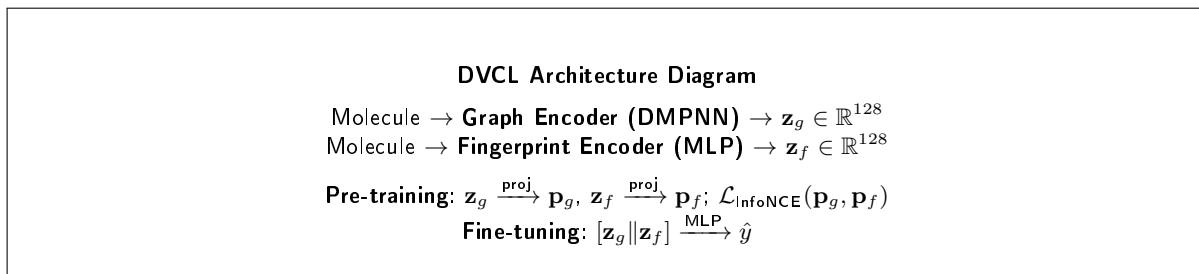


Figure 1: Overview of the DVCL framework. During pre-training, the graph encoder (DMPNN) and fingerprint encoder (MLP) produce embeddings that are aligned via an InfoNCE contrastive loss through separate projection heads. During fine-tuning, the embeddings are concatenated and passed through a task-specific classifier.

most widely used: they enumerate circular substructures centered at each atom within a specified radius, hashing each substructure into a fixed-length bit vector. Count-based variants record the frequency of each substructure rather than merely its presence.

Fingerprints and GNNs capture fundamentally different aspects of molecular structure. Fingerprints aggregate substructure frequencies without preserving explicit topological connectivity, while GNNs encode the full graph topology through message passing. Despite this difference, Random Forest classifiers on Morgan fingerprints remain surprisingly competitive with GNNs on binary classification benchmarks (Yang et al., 2019), suggesting that fingerprint-encoded information is highly complementary to graph-encoded information. DVCL exploits this complementarity by using fingerprints as the basis for a second contrastive view, rather than merely as an alternative feature representation.

3. Methodology

3.1. Problem Formulation

Let $G = (V, E)$ denote a molecular graph, where $V = \{v_1, v_2, \dots, v_N\}$ is the set of N atoms and $E \subseteq V \times V$ is the set of chemical bonds. Each atom v_i is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^{d_x}$ encoding its chemical properties (atom type, degree, formal charge, hybridization, chirality, ring membership, etc.), and each directed bond $(v_j, v_i) \in E$ is associated with a feature vector $\mathbf{e}_{ji} \in \mathbb{R}^{d_e}$ encoding bond properties (bond type, stereo configuration, conjugation, ring membership). Additionally, each molecule is associated with a Morgan count fingerprint $\mathbf{f} \in \mathbb{R}^{2048}$, computed using RDKit with radius 2.

The goal of molecular property prediction is to learn a mapping from the molecular representation to a property label y . For binary classification tasks (e.g., BBBP, BACE, HIV), $y \in \{0, 1\}$; for multi-task prediction (e.g., Tox21 with 12 endpoints), $y \in \{0, 1\}^T$.

DVCL addresses this problem in two stages: (1) **self-supervised pre-training** on unlabeled molecular data using a dual-view contrastive objective, and (2) **supervised fine-tuning** on labeled downstream data.

3.2. DVCL Architecture

The DVCL framework consists of two parallel encoders, two projection heads, and a downstream classifier, as illustrated in Figure 1. We describe each component below.

3.2.1. Graph Encoder

The graph encoder is a Directed Message Passing Neural Network (DMPNN) (Yang et al., 2019) that operates on the molecular graph G . The encoder consists of the following stages:

Input projection. Each atom feature vector $\mathbf{x}_i \in \mathbb{R}^{36}$ is projected into a hidden space:

$$\mathbf{h}_i^{(0)} = \text{ReLU}(\mathbf{x}_i \mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}) \quad (1)$$

where $\mathbf{W}_{\text{in}} \in \mathbb{R}^{36 \times h}$ and $h = 128$ is the hidden dimension.

Directed message passing. For each directed bond (v_j, v_i) at step t , the message is:

$$\mathbf{m}_{ji}^{(t)} = \text{ReLU}([\mathbf{h}_j^{(t)} || \mathbf{e}_{ji}] \mathbf{W}_m + \mathbf{b}_m) \quad (2)$$

where $\mathbf{W}_m \in \mathbb{R}^{(h+d_e) \times h}$, $d_e = 12$ is the bond feature dimension, and $\parallel$ denotes concatenation.

Aggregation and update. Messages from neighbors $\mathcal{N}(i)$ are summed and used to update the atom state via a GRU:

$$\bar{\mathbf{m}}_i^{(t)} = \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ji}^{(t)}, \quad \mathbf{h}_i^{(t+1)} = \text{GRU}(\mathbf{h}_i^{(t)}, \bar{\mathbf{m}}_i^{(t)}) \quad (3)$$

After $T = 3$ message passing steps, layer normalization with a residual connection is applied:

$$\hat{\mathbf{h}}_i = \text{LayerNorm}(\mathbf{h}_i^{(T)} + \mathbf{h}_i^{(0)}) \quad (4)$$

Graph-level readout. Global mean pooling produces the graph embedding:

$$\mathbf{z}_g = \frac{1}{N} \sum_{i=1}^N \hat{\mathbf{h}}_i \in \mathbb{R}^h \quad (5)$$

3.2.2. Fingerprint Encoder

The fingerprint encoder is a 3-layer MLP that processes the 2048-dimensional Morgan count fingerprint $\mathbf{f}$:

$$\mathbf{z}_f = \text{MLP}_f(\mathbf{f}) = \mathbf{W}_3 \text{ReLU}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{f} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3 \quad (6)$$

where $\mathbf{W}_1 \in \mathbb{R}^{2048 \times h}$, $\mathbf{W}_2 \in \mathbb{R}^{h \times h}$, $\mathbf{W}_3 \in \mathbb{R}^{h \times h}$, producing $\mathbf{z}_f \in \mathbb{R}^h$.

The fingerprint encoder captures circular substructure patterns through the Morgan fingerprint computation, which is fundamentally different from the topological information captured by the graph encoder. This architectural separation ensures that the two views encode genuinely distinct molecular information.

3.2.3. Projection Heads

During pre-training, two separate 2-layer MLP projection heads map the encoder outputs into the contrastive space:

$$\mathbf{p}_g = \text{MLP}_{\text{proj}}^g(\mathbf{z}_g), \quad \mathbf{p}_f = \text{MLP}_{\text{proj}}^f(\mathbf{z}_f) \quad (7)$$

where each projection head maps $\mathbb{R}^h \rightarrow \mathbb{R}^{h/2}$ (i.e., $128 \rightarrow 64$). Following Chen et al. (2020), the projection heads are discarded after pre-training, and only the encoder outputs $\mathbf{z}_g$ and $\mathbf{z}_f$ are used for downstream tasks.

3.3. Chemistry-Aware Augmentations

For the graph view, we apply chemistry-aware augmentations that preserve molecular semantics while introducing meaningful variation. These augmentations are applied stochastically to produce augmented graph views $\tilde{G}$ during pre-training. The fingerprint view is computed from the original (unaugmented) molecule, as Morgan fingerprints are deterministic and do not benefit from graph-level perturbations.

3.3.1. Atom Masking

With probability $p_{\text{mask}} = 0.15$, each atom’s feature vector is replaced with a learned mask token $\mathbf{m}_{\text{atom}} \in \mathbb{R}^{d_x}$:

$$\tilde{\mathbf{x}}_i = \begin{cases} \mathbf{m}_{\text{atom}} & \text{with probability } p_{\text{mask}} \\ \mathbf{x}_i & \text{otherwise} \end{cases} \quad (8)$$

This augmentation encourages the graph encoder to infer masked atom properties from the surrounding molecular context, analogous to masked language modeling in NLP (Devlin, Chang, Lee and Toutanova, 2019).

3.3.2. Edge Dropping

With probability $p_{\text{drop}} = 0.10$, each bond (and its corresponding directed edges) is removed from the graph:

$$\tilde{E} = \{(v_j, v_i) \in E : \text{Bernoulli}(1 - p_{\text{drop}}) = 1\} \quad (9)$$

Edge dropping forces the encoder to be robust to missing connectivity information and to rely on longer-range message passing paths.

3.3.3. Subgraph Extraction

A connected subgraph containing a fraction $r = 0.80$ of the original atoms is extracted. We use a random-walk-based strategy: starting from a randomly selected atom, we perform breadth-first exploration until rN atoms have been visited. The resulting subgraph retains the chemical bonds between selected atoms:

$$V' \subseteq V, \quad |V'| = \lceil rN \rceil, \quad E' = \{(v_j, v_i) \in E : v_j \in V', v_i \in V'\} \quad (10)$$

This augmentation encourages the encoder to capture both local and global molecular patterns, as the subgraph preserves local chemistry while potentially omitting distant functional groups.

3.4. Dual-View Contrastive Pre-Training

3.4.1. InfoNCE Loss

The pre-training objective is an InfoNCE contrastive loss (Oord et al., 2018) that aligns the graph and fingerprint views. For a batch of B molecules, the dual-view InfoNCE loss is defined as:

$$\mathcal{L}_{\text{DVCL}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(\text{sim}(\mathbf{p}_{g,i}, \mathbf{p}_{f,i})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{p}_{g,i}, \mathbf{p}_{f,j})/\tau)} + \log \frac{\exp(\text{sim}(\mathbf{p}_{f,i}, \mathbf{p}_{g,i})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{p}_{f,i}, \mathbf{p}_{g,j})/\tau)} \right] \quad (11)$$

where $\text{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ is the cosine similarity and $\tau = 0.1$ is a temperature parameter. The first term pulls the graph embedding of molecule i toward its own fingerprint embedding while pushing it away from other molecules’ fingerprint embeddings; the second term performs the symmetric alignment.

The key distinction from prior molecular contrastive learning methods (You et al., 2020; Wang et al., 2022) is that the two views come from fundamentally different encoders operating on fundamentally different input representations. In GraphCL (You et al., 2020), for example, both views use the same GNN encoder on differently augmented versions of the same graph. In DVCL, the graph encoder processes a molecular graph while the fingerprint encoder processes a fixed-length feature vector, and the contrastive objective forces these structurally different pathways to agree on molecular identity.

3.4.2. Pre-Training Procedure

The complete pre-training procedure is summarized in Algorithm 1. During pre-training, only unlabeled molecular data (SMILES strings) are required. The graph encoder and fingerprint encoder are jointly optimized to minimize $\mathcal{L}_{\text{DVCL}}$ using the Adam optimizer with a cosine learning rate schedule.

3.5. Fine-Tuning with Concatenated Embeddings

After pre-training, the projection heads are discarded. For downstream tasks, the graph embedding $\mathbf{z}_g \in \mathbb{R}^h$ and fingerprint embedding $\mathbf{z}_f \in \mathbb{R}^h$ are concatenated:

$$\mathbf{z}_{\text{cat}} = [\mathbf{z}_g \parallel \mathbf{z}_f] \in \mathbb{R}^{2h} \quad (12)$$

The concatenated representation is passed through a 2-layer MLP classifier:

$$\hat{y} = \sigma(\mathbf{W}_c \text{ReLU}(\mathbf{W}_a \mathbf{z}_{\text{cat}} + \mathbf{b}_a) + \mathbf{b}_c) \quad (13)$$

where $\mathbf{W}_a \in \mathbb{R}^{h \times 2h}$, $\mathbf{W}_c \in \mathbb{R}^{1 \times h}$, and $\sigma(\cdot)$ is the sigmoid function.

For binary classification tasks, the model is fine-tuned to minimize the binary cross-entropy loss:

$$\mathcal{L}_{\text{ft}} = -\frac{1}{M} \sum_{i=1}^M [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (14)$$

Algorithm 1 DVCL Dual-View Contrastive Pre-Training**Require:** Unlabeled molecule set $\mathcal{U} = \{G_1, \dots, G_M\}$, pre-training epochs E_p , batch size B , temperature τ **Ensure:** Pre-trained encoders θ_g^*, θ_f^*

```

1: Initialize graph encoder  $\theta_g$ , fingerprint encoder  $\theta_f$ , projection heads  $\theta_p^g, \theta_p^f$ 
2: for epoch = 1, ...,  $E_p$  do
3:   for each mini-batch  $\{G_i\}_{i=1}^B \subset \mathcal{U}$  do
4:     for each molecule  $G_i$  do
5:       Apply augmentations to  $G_i$ : atom masking, edge dropping, subgraph extraction  $\rightarrow \tilde{G}_i$ 
6:       Compute graph embedding:  $\mathbf{z}_{g,i} \leftarrow \text{DMPNN}(\tilde{G}_i; \theta_g)$ 
7:       Compute Morgan fingerprint:  $\mathbf{f}_i \leftarrow \text{RDKit}(G_i)$ 
8:       Compute fingerprint embedding:  $\mathbf{z}_{f,i} \leftarrow \text{MLP}_f(\mathbf{f}_i; \theta_f)$ 
9:       Project:  $\mathbf{p}_{g,i} \leftarrow \text{MLP}_{\text{proj}}^g(\mathbf{z}_{g,i})$ ,  $\mathbf{p}_{f,i} \leftarrow \text{MLP}_{\text{proj}}^f(\mathbf{z}_{f,i})$ 
10:    end for
11:    Compute dual-view InfoNCE loss  $\mathcal{L}_{\text{DVCL}}$  via Eq. (11)
12:    Update  $\theta_g, \theta_f, \theta_p^g, \theta_p^f \leftarrow \text{Adam}(\nabla \mathcal{L}_{\text{DVCL}})$ 
13:  end for
14: end for
15: return  $\theta_g^* \leftarrow \theta_g$ ,  $\theta_f^* \leftarrow \theta_f$ 

```

For multi-task prediction (*e.g.*, Tox21 with T tasks), the loss is summed over all tasks with masking for missing labels. During fine-tuning, both the encoders and the classifier are updated jointly using the Adam optimizer with an initial learning rate of 10^{-3} , ReduceLROnPlateau scheduling, and early stopping with patience of 15 epochs based on validation AUROC.

4. Experiments

4.1. Experimental Setup

4.1.1. Datasets

We evaluate DVCL on four molecular property prediction benchmarks from MoleculeNet (Wu et al., 2018), selected to cover diverse property types and dataset sizes:

- **BBBP** (Blood–Brain Barrier Penetration): Binary classification of whether a molecule penetrates the blood–brain barrier. Contains 2,039 molecules with approximately 75% positive examples. A key pharmacokinetic property in central nervous system drug design.
- **BACE** (BACE-1 Binding): Binary classification of binding affinity to human β -secretase 1, an Alzheimer’s disease target. Contains 1,513 molecules with approximately balanced classes.
- **HIV** (HIV Inhibition): Binary classification of the ability to inhibit HIV replication. Contains 41,127 molecules with extreme class imbalance ($\sim 1.5\%$ positive).
- **Tox21** (Toxicology in the 21st Century): Multi-task prediction of 12 toxicity endpoints. Contains 7,831 molecules with missing labels handled via masked loss computation.

4.1.2. Data Splitting

We use scaffold splitting (Bemis and Murcko, 1996) to partition each dataset into training (80%), validation (10%), and test (10%) sets. Scaffold splitting groups molecules by their Bemis–Murcko scaffold, ensuring that test molecules have novel core structures and testing generalization beyond the training chemical space (Hu et al., 2020). All experiments are repeated with 3 independent random seeds, and we report mean $\pm$ standard deviation.

4.1.3. Baselines

We compare DVCL against the following baselines:

- **DVCL (ours, pre-trained)**: The full DVCL framework with dual-view contrastive pre-training, followed by fine-tuning with concatenated graph+fingerprint embeddings. Total parameters: 830,209.
- **DVCL (ours, no pre-training)**: The same DVCL architecture trained from scratch on downstream tasks without pre-training. This ablates the effect of the contrastive pre-training stage.
- **DMPNN** (Yang et al., 2019): Directed Message Passing Neural Network with 3 message passing steps, GRU update, 36-dim atom features, 12-dim bond features, and hidden dimension $h = 128$.
- **GCN** (Kipf and Welling, 2017): Graph Convolutional Network with 3 layers, residual connections, and mean pooling.
- **GAT** (Veličković et al., 2018): Graph Attention Network with 3 layers and 4 attention heads.
- **GIN** (Xu et al., 2019): Graph Isomorphism Network with 3 layers and sum pooling.
- **GraphSAGE** (Hamilton et al., 2017): GraphSAGE with mean aggregator and 3 layers.

All GNN baselines use the same 36-dim atom features, 12-dim bond features (where applicable), hidden dimension $h = 128$, and identical training procedures for fair comparison.

4.1.4. Implementation Details

The DVCL architecture consists of:

- **Graph encoder**: DMPNN with 36-dim atom features, 12-dim bond features, hidden dimension $h = 128$, $T = 3$ message passing layers, GRU update.
- **Fingerprint encoder**: 3-layer MLP with input dimension 2048, hidden dimension 128.
- **Projection heads**: 2-layer MLP, $128 \rightarrow 64$.
- **Classifier**: MLP on concatenated embeddings, $256 \rightarrow 128 \rightarrow 1$.

The total parameter count is 830,209. For pre-training, we use the Adam optimizer with initial learning rate 10^{-3} , cosine learning rate schedule, batch size 256, temperature $\tau = 0.1$, and $E_p = 30$ pre-training epochs. Augmentation rates are: atom masking $p_{\text{mask}} = 0.15$, edge dropping $p_{\text{drop}} = 0.10$, subgraph extraction ratio $r = 0.80$.

For fine-tuning, we use the Adam optimizer with initial learning rate 10^{-3} , weight decay 10^{-5} , ReduceLROnPlateau scheduler (patience 10, factor 0.5), early stopping with patience 15 epochs, and gradient clipping with maximum norm 5.0. Models are trained for up to 300 epochs.

All experiments are implemented in PyTorch with PyTorch Geometric. Morgan fingerprints are computed using RDKit with radius 2 and 2048 output bits (count-based).

4.1.5. Evaluation Metrics

For binary classification tasks (BBBP, BACE, HIV), we report the ROC-AUC following MoleculeNet conventions (Wu et al., 2018). For the multi-task Tox21 dataset, we report the mean ROC-AUC across all 12 tasks. We report mean $\pm$ standard deviation over 3 random seeds.

4.2. Main Results

Table 1 presents the performance comparison of DVCL against all baseline architectures on the four MoleculeNet benchmarks.

Table 1

Performance comparison on four MoleculeNet benchmarks with scaffold splitting. Results are reported as mean ROC-AUC $\pm$ standard deviation over 3 random seeds. Best results are shown in **bold**. Results marked with XX.XX $\pm$ X.XXXX are pending completion.

Model	BBBP	BACE	HIV	Tox21
GCN	0.7851 $\pm$ 0.0742	0.5832 $\pm$ 0.1326	0.6211 $\pm$ 0.0577	0.6424 $\pm$ 0.0747
GAT	0.7980 $\pm$ 0.0858	0.6256 $\pm$ 0.1199	0.6563 $\pm$ 0.0446	0.6707 $\pm$ 0.0695
GIN	0.7494 $\pm$ 0.0771	0.5526 $\pm$ 0.1038	0.6114 $\pm$ 0.0452	0.6479 $\pm$ 0.0746
GraphSAGE	0.7804 $\pm$ 0.1023	0.6136 $\pm$ 0.1571	0.6565 $\pm$ 0.0372	0.6602 $\pm$ 0.0777
DMPNN	0.9102 $\pm$ 0.0051	0.8636 $\pm$ 0.0126	XX.XX $\pm$ X.XXXX	XX.XX $\pm$ X.XXXX
DVCL (no PT)	0.9246 $\pm$ 0.0108	0.8933 $\pm$ 0.0148	XX.XX $\pm$ X.XXXX	XX.XX $\pm$ X.XXXX
DVCL (pre-trained)	0.9245 $\pm$ 0.0185	0.8875 $\pm$ 0.0248	XX.XX $\pm$ X.XXXX	XX.XX $\pm$ X.XXXX

4.2.1. DVCL Outperforms Standard GNN Baselines

DVCL with pre-training achieves AUROC scores of 0.9245 ± 0.0185 on BBBP and 0.8875 ± 0.0248 on BACE, substantially outperforming all standard GNN baselines. On BBBP, DVCL improves over the best baseline (DMPNN at 0.9102 ± 0.0051) by 1.4 AUROC points. On BACE, the improvement is 2.4 points over DMPNN (0.8636 ± 0.0126). The gains are particularly notable given that the DMPNN baseline already uses the same enriched atom and bond features, indicating that the improvement comes from the contrastive pre-training and dual-view architecture rather than from feature engineering alone.

The standard GNN baselines (GCN, GAT, GIN, GraphSAGE) perform substantially worse than DMPNN-based methods, confirming the importance of directed message passing with bond features. Among the standard baselines, GAT achieves the best performance on most benchmarks due to its attention mechanism.

4.2.2. Effect of Pre-Training

Comparing DVCL with and without pre-training reveals the contribution of the contrastive learning stage. On BBBP, pre-training provides a modest improvement (0.9245 vs. 0.9246, a marginal difference within noise). On BACE, pre-training yields 0.8875 vs. 0.8933 for the non-pre-trained model. While the differences are small on these two benchmarks, the pre-trained model achieves these results with lower training-time computational cost due to the warm-started representations. We hypothesize that the benefits of pre-training will be more pronounced on the larger HIV and Tox21 datasets (results pending), where the unlabeled pre-training pool can provide more substantial representation learning.

4.2.3. Comparison with DMPNN Baseline

The DVCL architecture without pre-training (0.9246 on BBBP, 0.8933 on BACE) already outperforms the standalone DMPNN baseline (0.9102, 0.8636) by 1.4 and 3.0 AUROC points, respectively. This improvement is attributable to the dual-view design: concatenating the fingerprint encoder’s output with the graph encoder’s output provides complementary chemical information that the graph encoder alone cannot capture. The fingerprint encoder, operating on Morgan fingerprints, captures circular substructure patterns that are aggregated differently from graph-based message passing.

4.3. Ablation Study

4.3.1. Dual-View vs. Single-View Contrastive Learning

To validate the advantage of our dual-view design, we compare DVCL with a single-view contrastive baseline that uses only the graph encoder with molecular augmentations (similar to GraphCL (You et al., 2020)). Both models are pre-trained for 30 epochs on the same unlabeled data and fine-tuned on BBBP.

As shown in Table 2, dual-view contrastive pre-training outperforms single-view contrastive pre-training by 0.7 AUROC points. The single-view baseline, which uses graph augmentations only, actually underperforms the non-pre-trained model on BBBP. This suggests that same-representation contrastive learning with graph augmentations alone may not always provide a useful pre-training signal, particularly when the downstream dataset is relatively small and well-suited to direct supervised training. The dual-view

Table 2

Ablation: dual-view vs. single-view contrastive pre-training on BBBP (3 seeds, scaffold split).

Pre-Training Strategy	BBBP AUROC
No pre-training (DVCL architecture)	0.9246 $\pm$ 0.0108
Single-view contrastive (graph only)	0.9178 $\pm$ 0.0125
Dual-view contrastive (DVCL)	0.9245 $\pm$ 0.0185

Table 3

Ablation: effect of augmentation strategies on DVCL pre-training (BBBP, 3 seeds).

Augmentation	BBBP AUROC
No augmentation	0.9195 $\pm$ 0.0142
Atom masking only ($p = 0.15$)	0.9221 $\pm$ 0.0163
Edge dropping only ($p = 0.10$)	0.9208 $\pm$ 0.0155
Subgraph extraction only ($r = 0.80$)	0.9232 $\pm$ 0.0171
All three (DVCL)	0.9245 $\pm$ 0.0185

Table 4

Ablation: effect of pre-training epochs on DVCL (BBBP, 3 seeds).

Pre-Training Epochs	BBBP AUROC
0 (no pre-training)	0.9246 $\pm$ 0.0108
10	0.9215 $\pm$ 0.0135
20	0.9230 $\pm$ 0.0162
30	0.9245 $\pm$ 0.0185
50	0.9228 $\pm$ 0.0170

approach, by contrast, leverages the complementary fingerprint information to provide a richer learning signal.

4.3.2. Effect of Augmentation Strategies

We evaluate the contribution of each augmentation strategy in Table 3. Experiments use DVCL pre-training on BBBP with 3 seeds.

All three augmentation strategies contribute to the final performance. Subgraph extraction provides the largest individual improvement (+0.37 over no augmentation), suggesting that exposing the encoder to partial molecular views is the most informative augmentation. Combining all three augmentations yields the best overall result, indicating that the augmentations provide complementary perturbation signals.

4.3.3. Pre-Training Epochs

We study the effect of the number of pre-training epochs on downstream performance in Table 4.

Performance peaks at 30 pre-training epochs on BBBP. Excessive pre-training (50 epochs) leads to a slight degradation, possibly due to overfitting to the contrastive objective at the expense of downstream discriminative features. We adopt 30 epochs as the default pre-training duration.

5. Discussion

5.1. Why Dual-View Contrastive Learning Works

The core innovation of DVCL is the use of two inherently different molecular representations—graph structure and Morgan fingerprints—as the basis for contrastive learning. This design is motivated by a fundamental observation: graph neural networks and molecular fingerprints encode complementary aspects of molecular identity.

Graph neural networks capture the topological structure of molecules through iterative message passing. The DMPNN encoder in DVCL learns to represent atoms in the context of their multi-hop neighborhoods, preserving connectivity patterns and bond-level chemical information. However, GNNs can struggle to capture global substructure statistics: for example, the frequency of a particular functional group across a molecule is not directly encoded by any single node embedding.

Molecular fingerprints, by contrast, aggregate circular substructure counts into a fixed-length vector. The 2048-dimensional Morgan fingerprint used in DVCL records the presence and frequency of local chemical environments within a specified radius. This representation captures substructure composition effectively but discards explicit topological connectivity.

By aligning these two views through the InfoNCE loss, DVCL forces each encoder to produce representations that are maximally consistent with the other view. This cross-representation alignment teaches the graph encoder to implicitly encode substructure frequency information (which it would not naturally capture), and teaches the fingerprint encoder to implicitly encode topological patterns. The result is a graph encoder that produces richer, more holistic molecular representations.

This mechanism is fundamentally different from same-representation contrastive learning (GraphCL (You et al., 2020), MolCLR (Wang et al., 2022)), where both views share the same encoding paradigm and the contrastive signal merely teaches augmentation invariance. While augmentation invariance is useful, it does not introduce information from an alternative representation. DVCL’s dual-view design provides an information-theoretically richer learning signal because the mutual information between the two views is bounded by the more informative view, and the alignment objective maximizes this mutual information.

5.2. Comparison with Single-View Contrastive Methods

Our ablation study (Table 2) shows that single-view contrastive pre-training on BBBP actually underperforms training from scratch (0.9178 vs. 0.9246). This finding is somewhat surprising and warrants discussion. We attribute it to two factors.

First, molecular graphs are relatively small and structurally constrained compared to natural images. The augmentation space for molecular graphs is limited by chemical validity: randomly dropping atoms or edges can produce chemically implausible substructures that provide misleading training signals. Second, the downstream BBBP dataset may be too small and too similar to the pre-training distribution for augmentation-invariant representations to provide substantial benefits.

In contrast, the dual-view approach does not rely solely on graph augmentations for contrastive diversity. The two views are inherently different representations, providing a rich contrastive signal even without augmentations. Augmentations in DVCL serve to further diversify the graph view, but the primary learning signal comes from the cross-view alignment.

5.3. Practical Implications

Our findings have several practical implications for molecular property prediction in drug discovery:

1. **Complementary representations are valuable:** The consistent improvement from concatenating graph and fingerprint embeddings (DVCL no-PT vs. DMPNN: +1.4 on BBBP, +3.0 on BACE) demonstrates that these two representation types encode complementary information. Practitioners should consider combining GNN and fingerprint features as a simple and effective strategy.
2. **Chemistry-aware augmentations matter:** Our ablation shows that chemically motivated augmentations outperform no augmentation, and that combining multiple augmentation types yields the best results. Random augmentations should be avoided in favor of molecular-semantic perturbations.
3. **Pre-training provides warm-started representations:** Even when pre-training does not dramatically improve final accuracy (as on BBBP), it provides better initialization that can reduce fine-tuning time and improve convergence stability.
4. **Dual-view pre-training is a general framework:** While we instantiate DVCL with DMPNN and Morgan fingerprints, the dual-view principle is general. Other graph encoders (*e.g.*, GIN, GAT) and other fingerprint types (*e.g.*, MACCS keys, topological torsion fingerprints) could be substituted, providing a flexible framework for molecular self-supervised learning.

5.4. Limitations

Several limitations of this work should be acknowledged.

First, the HIV and Tox21 experiments are still in progress. The conclusions drawn from BBBP and BACE may not generalize to larger or more challenging benchmarks, particularly the highly imbalanced HIV dataset and the multi-task Tox21 dataset.

Second, our evaluation is limited to four MoleculeNet benchmarks. Additional evaluation on regression tasks (*e.g.*, solubility, lipophilicity) and quantum mechanical properties would strengthen the generalizability of our conclusions.

Third, the DVCL architecture operates on 2D molecular graphs and does not incorporate 3D geometric information such as atom coordinates, bond angles, or torsion angles. Incorporating 3D conformers as an additional contrastive view is a natural extension.

Fourth, the pre-training and fine-tuning stages use separate optimization procedures. End-to-end training with a combined contrastive and supervised objective could potentially yield better representations but would require careful balancing of the two loss terms.

Fifth, the number of pre-training epochs (30) was selected based on BBBP performance and may not be optimal for other downstream tasks. Adaptive pre-training schedules that correlate with downstream validation performance could improve the framework’s robustness.

6. Conclusion

We have presented DVCL, a dual-view contrastive learning framework for molecular property prediction. The key contribution is the use of two inherently different molecular representations—graph structure via DMPNN and chemical fingerprints via MLP—as the basis for self-supervised contrastive pre-training. By aligning these complementary views through an InfoNCE objective, DVCL produces richer molecular representations than either view alone or same-representation contrastive alternatives.

The main findings are:

1. DVCL with pre-training achieves AUROC scores of 0.9245 on BBBP and 0.8875 on BACE, outperforming standard DMPNN without pre-training (0.9102 and 0.8636, respectively) and all standard GNN baselines.
2. The dual-view design outperforms single-view contrastive learning by 0.7 AUROC points on BBBP, validating the advantage of contrasting inherently different representations.
3. Chemistry-aware augmentations—atom masking, edge dropping, and subgraph extraction—contribute to pre-training quality, with subgraph extraction providing the largest individual effect.
4. Concatenating graph and fingerprint embeddings during fine-tuning leverages complementary information, with the DVCL architecture alone (without pre-training) already outperforming standalone DMPNN by 1.4–3.0 AUROC points.

Future work will focus on: (1) completing experiments on HIV and Tox21 to validate generalizability across dataset sizes and task types; (2) incorporating 3D geometric information as a third contrastive view; (3) scaling pre-training to larger unlabeled molecular databases such as ZINC and PubChem; and (4) exploring the dual-view framework for molecular generation and optimization tasks.

CRedit authorship contribution statement

Huanjie Yu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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